Suman Dangal

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EDUCATION

Springfield College, Springfield, MA *Expected May 2028*

B.S. in Computer Science and Mathematics | Honors Program | GPA: 3.986/4.000 | Dean's List, all terms

Relevant Coursework: Deep Learning Fundamentals I & II, Linear Algebra, Discrete Mathematics, Probability & Statistics, Advanced Python Programming, Data Structures (C++)

PUBLICATIONS & RESEARCH

"Enhancing Diabetes Classification with Echo State Networks: Parallel Architectures and Feature Engineering Approaches" *Green Cities Conference, Jun 2025*

- Achieved 100% classification accuracy on the Pima Indians Diabetes Dataset, surpassing every prior published benchmark, by engineering second-order feature interactions (e.g. glucose×BMI) into a reservoir-tuned Echo State Network
- Awarded Best Paper (Knowledge Management) and Best Presentation (Undergraduate); featured in Springfield College's achievement publication (Laurels, Vol. 12 Issue 2, Fall 2025)

"Analysis of Pathological Evidence in Microscopic Images using CNN and Deep Learning Methods with Telemedicine Facilities" (Dristi AI) *APCYS 2023*

- Detected malaria parasites and sickle cells and counted RBCs, WBCs, and platelets from blood-smear microscopy images by training a TinyYOLO/CNN pipeline via transfer learning
- Backed a low-cost Raspberry Pi deployment for rural diagnostics with a signed clinical-partner MOU for AI-output verification; awarded Silver Medal, Computer Science category

PROJECTS

Financial Deal Agent, Independent Research *2026 – Present*

- Improved provenance traceability in financial-document reasoning over flat vector retrieval by building page-indexed retrieval across a hierarchical knowledge graph (company to filing to section to claim)

Graph Theory Visualizer, Independent Research *2024 – Present*

- Investigating equitable graph coloring (Hajnal-Szemeredi framework), profiling runtime and memory trade-offs across data structures and orderings

Course RAG Study Assistant, Personal Tool *2025 – Present*

- Built a retrieval-augmented generation tool over course material to help read and practice for Deep Learning Fundamentals I & II

EXPERIENCE

Co-founder & CTO, Saalik Tech Pvt. Ltd, Kathmandu, Nepal *May 2023 – Present*

- Delivered a proof-of-concept object-detection model at 92% accuracy across 150+ cultural artifacts by leading AI-driven feasibility studies at Saalik Tech

Teaching Assistant, Data Structures & Algorithms (C++), Springfield College *Jan 2026 – Present*

- Lead C++ labs for 30+ students on pointers, memory management, and data structures

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow/Keras, OpenCV, YOLO (transfer learning), Reservoir Computing (ESN), RAG & graph-based retrieval

Languages & Tools: Python, C++, Go, SQL, Git/GitHub, Linux